



Testosterone for Hypogonadism in Males: Amber Recommendation Guideline for Primary Care

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TESTOSTERONE FOR HYPOGONADISM IN MALES

Amber Recommendation Guideline for Primary Care

This protocol provides prescribing and monitoring guidance for testosterone therapy for management of hypogonadism in adult cis male patients and does not cover any use of testosterone for treatment of children or of trans male patients. It should be read in conjunction with the Summary of Product Characteristics (SPC) available on www.medicines.org.uk/emc and the [BNF](#).

BACKGROUND FOR USE

Low testosterone (hypogonadism) is associated with a number of symptoms and signs. Symptoms include low libido and, if severe, erectile dysfunction in particular loss of nocturnal spontaneous erections, loss of energy, fatigue, depression, decrease in cognitive abilities and irritability. Low testosterone can also have negative effects on bone mass, which results in a significant risk of osteoporosis in hypogonadal men. Progressive reductions in muscle mass and anaemia may also be a consequence of low testosterone in men

The causes for low testosterone may be either primary, i.e. gonadal failure, or secondary to hypothalamic or pituitary disease. **It is essential to establish a cause before starting treatment**

Fertility intentions should always be discussed before starting Testosterone replacement. Testosterone replacement suppresses spermatogenesis.

Who should be assessed for hypogonadism:

- Those men presenting with symptoms compatible with hypogonadism as described above
- Men with hypothalamic or pituitary conditions
- Osteoporosis or low trauma fracture
- Erectile dysfunction (if other symptoms are suggestive of low testosterone)
- Type 2 diabetes in presence of symptoms of hypogonadism
- Men with infertility
- HIV associated with weight loss
- Chronic glucocorticoid therapy
- Chronic opioid therapy associated with symptoms of hypogonadism
- End-stage renal disease and maintenance haemodialysis
- Unexplained anaemia
- Consideration should be given to routinely asking men if they have any symptoms of hypogonadism especially those at high risk. These include men with diabetes, osteoporosis (fragility fractures), chronic opiate therapy, cardiovascular disease, erectile dysfunction and depression.

Diagnosis

It is advised that blood collection for total testosterone measurement be taken at 0900h in the morning. (Testosterone has a circadian rhythm - levels highest at 0600 - 0800h falling to a low at 1800 - 2000h.)

In the diagnosis of hypogonadism, two morning testosterone levels at least one week apart should be taken. There is no clear cut-off below which a diagnosis is made. The testosterone level has to be taken into context with the symptoms and underlying disease.

- <8 nmol/L is consistent with a diagnosis of hypogonadism.
- $8 - 12$ nmol/L could represent hypogonadism and patients may be considered for a trial of testosterone replacement therapy (TRT). Assessment should be done in secondary care and includes clinical history, calculation of free testosterone and investigation for evidence of structural pituitary or gonadal disease. Hypogonadism is more likely to be present with a testosterone <10.4 nmol/L.
- >12 nmol/L subject is not hypogonadal.

GP responsibilities

- Initial screening for hypogonadism in appropriate patients

- LH, FSH and prolactin to establish whether primary or secondary
- Baseline FBC, PSA, LFTs
- Refer to secondary care for diagnosis and for further investigations to establish cause
- Accept handover of prescribing responsibility or raise concerns
- Trough testosterone annually if on Nebido®; random testosterone (ideally 4 hours after gel application) annually if on testosterone gel. Dose change in testosterone is unlikely to be necessary once the patient is discharged on stable dose of testosterone (see below)
- FBC annually. Seek endocrine advice if hematocrit or hemoglobin raised
- Annual PSA

Consultant responsibilities

- Make diagnosis of hypogonadism and establish cause
- Discuss and manage fertility intentions/issues
- Baseline DEXA scan if appropriate
- Prior to testosterone initiation, all patients must undergo a detailed examination in order to exclude a risk of pre-existing prostatic cancer: PSA, ask about urinary symptoms, DRE if high risk
Advise GP as to type of testosterone replacement
- Review patient until stabilised on preferred Testosterone replacement product
- Bloods FBC, LFT, PSA, Ca until stabilised on preferred product usually this is at 3 months.
- Some patients, e.g. Klinefelter's, pituitary disorders, may remain under secondary care
- Some patients once stabilised on testosterone will be discharged to primary care

TREATMENT

The form of testosterone replacement is patient choice. Gels are used when initiating testosterone therapy as they are easily reversible and dose can be adjusted in the event of adverse reactions. Patients may choose to switch to injections for long term therapy. If injections are preferred then the first line is Nebido. Short acting intramuscular preparations of Testosterone esters are also available

Testosterone gels

Prescribed dose may be 20-60 mg /day titrated to target testosterone concentration. Testosterone concentration should be taken 2-6 hours after applying gel. Aim for serum testosterone concentration in the middle of the normal range

- Testogel: sachet 50 mg: useful if required dose is 50 mg once daily.
- Testogel pump: 16.2 mg/g: 20.25mg per actuation: useful for accurate administration of other doses
- Tostran 2% pump: 10 mg per pump actuation: useful for accurate administration of other doses
- Testavan pump: 23 mg per actuation: Novel hands free applicator to reduce secondary transmission (most cost effective product – Nov 23)

Tostran® gel pump 20 mg/g (2%)		Testavan® gel pump 20 mg/g (2%)		Testogel® 50 mg/sachet		Testogel® 40.5 mg/sachet		Testogel pump 16.2 mg/g (1%)	
20 mg	= 2 pumps	23 mg	= 1 pump	25 mg	= ½ sachet	20.25 mg	= ½ sachet	20.25 mg	= 1 pump
30 mg	= 3 pumps								
40 mg	= 4 pumps	46 mg	= 2 pumps			40.5 mg	= 1 sachet	40.5 mg	= 2 pumps
50 mg	= 5 pumps			50 mg	= 1 sachet				
60 mg	= 6 pumps					60.75 mg	= 1½ sachets	60.75 mg	= 3 pumps
70 mg	= 7 pumps	69 mg	= 3 pumps	75 mg	= 1½ sachets				
80 mg	= 8 pumps					81 mg	= 2 sachets	81 mg	= 4 pumps

Long acting Testosterone undecanoate Nebido® administered intra-muscularly lasting up to 3 months. The initial injection should be followed by a second at 6 weeks, and subsequent injections every 10 - 14 weeks until the patient is on a stable dose. Dosage interval is based on trough

testosterone concentration taken in the week before next injection due. Aim trough testosterone in lower end of normal range

Sustanon 250 mg/ml IM every 4 weeks. Potentially unpleasant peak and trough symptoms with higher risk of high haematocrit. Monitor with trough testosterone levels in few days before injection due and aim for lower end normal range. Not recommended for new patients but can be continued in existing patients if needed.

Testosterone Enanthate 150-200mg IM every 2 weeks or 50-100mg IM or S/C once a week. When administered S/C can be potentially be self-injected. Monitoring with trough Testosterone concentrations aiming for lower end of normal range. Potentially unpleasant peak and trough symptoms. Mainly reserved for patients who want to self-inject.

CONTRAINDICATIONS

- Hypersensitivity
 - Hypercalcaemia
 - In cases of known or suspected prostatic cancer or breast carcinoma
 - In cases of known hypersensitivity to the active substance or any of the excipients
 - Testosterone replacement is not indicated for use in children and has not been evaluated clinically in males under 18 years of age
 - Past or present liver tumours (Nebido®)
- For comprehensive information see relevant SPCs

PRECAUTIONS

Prior to testosterone initiation, all patients must undergo a detailed examination in order to exclude a risk of pre-existing prostatic cancer: PSA, ask about urinary symptoms, DRE if high risk. (see specialist responsibilities). Androgens may accelerate the progression of sub-clinical prostatic cancer and benign prostatic hyperplasia.

Due to variability in laboratory values, all measures of testosterone should be carried out in the same laboratory.

In patients suffering from severe cardiac, hepatic or renal insufficiency or ischaemic heart disease, treatment with testosterone may cause severe complications characterised by oedema with or without congestive cardiac failure. In such cases, treatment must be stopped immediately. In addition, diuretic therapy may be required.

Testosterone gels should be used with caution in cancer patients at risk of hypercalcaemia (and associated hypercalciuria), due to bone metastases. Regular monitoring of the serum levels of calcium in these patients is recommended.

Testosterone gels should also be used with caution in ischaemic heart disease, hypertension, thrombophilia, epilepsy and migraine.

DRUG INTERACTIONS (See SPC for full list <http://www.medicines.org.uk/emc/>.)

Oral coumarins-based anticoagulant	Increased monitoring of international normalised ratio (INR) recommended particularly when androgens started or stopped.
Corticosteroids/adrenocorticotrophic hormone (ACTH)	Increased risk of developing oedema. Co-administer with caution.
Insulin	Improved insulin sensitivity may occur in patients treated with androgens who achieve normal testosterone plasma concentrations following replacement therapy.

MONITORING GUIDE FOR PRIMARY CARE:

Conditions requiring dose adjustment:

- Patient experiencing side effects

- Raised PSA or haematocrit
- Raised or low/out of range trough testosterone level for patients on IM testosterone (aim is for trough level to be in the lower third of normal range): dose interval will need adjusting as above
- Raised or low/out of range serum testosterone taken 2-6 hours after gel application: may need adjustment of gel dose.
- New co-existing diagnosis e.g. Prostate cancer. In this case testosterone would most likely be stopped pending discussion with Urology (refer to Specialist)

Monitoring: guidance and actions regarding PSA and FBC monitoring (at 3 months and annually)

PSA:

If PSA becomes raised above the normal range (see table 1), please inform Specialist and consider referral to Urology

Normal age-specific range for PSA (Table 1)

Age Range	40 - 49	50 - 59	60 - 69	70 – 79	≥ 80	
Threshold for Urology referral based on PSA alone	> 2.5	*NEW* > 3.5	*NEW* > 4.5	*NEW* > 6.5	*NEW* Normal fitness	*NEW* > 20.0
					Very fit	>10
Halve the threshold if on 5α Reductase Inhibitors	> 1.25 (on 5αRI)	> 1.5 (on 5αRI)	> 2.0 (on 5αRI)	>2.5 (on 5αRI)	> 5.0 (on 5αRI)	

FBC - Haematocrit/ Hct monitoring on testosterone replacement:

- Monitoring should be performed: baseline, 3 months, 6 months (specialist) and minimum of annually thereafter until stopped treatment.
- If elevated:
 - Hct > 0.52 GP to repeat bloods in 1-2 weeks to assess if transient, and then reassess monitoring.
 - Hct > 0.54 GP to use A+G or referral to Endocrine Team.

REFERENCES

Society for Endocrinology Guidelines for Testosterone Replacement Therapy in Male Hypogonadism. Jayasena et al. Clinical Endocrinology 2022;96:200-219